

Finance AI Chatbot

Built on Google Dialogflow NLP

MBA PROJECT PRESENTATION

9+

Intents Configured

1

Custom Entity

3

Financial Modules

Loan Eligibility • EMI Calculation • Investment Advice

Problem Statement

Why does personal finance need an AI assistant?

Financial Literacy Gap

Millions of Indians lack access to basic financial guidance on loans and investments.

Manual & Time-Consuming

Visiting banks or financial advisors for EMI/loan queries is slow and inconvenient.

No 24×7 Support

Traditional banking helplines are unavailable outside business hours.

Complex Calculations

EMI formulas and loan eligibility rules confuse the average user.

One-Size Advice

Generic financial advice ignores individual income, risk appetite, and goals.

Objectives

What this chatbot is designed to achieve



Loan Eligibility

Instantly evaluate if a user qualifies for a home/personal loan based on income and loan amount.



EMI Estimation

Calculate monthly installments using the standard banking EMI formula (P, r, n).



Investment Guidance

Provide risk-level-based investment recommendations tailored to the user's profile.



NLP-Powered UX

Leverage Google Dialogflow's NLP to understand natural language financial queries.



Always Available

Deliver 24x7 financial assistance without requiring a human advisor.



Financial Literacy

Educate users on financial concepts through conversational AI interaction.

AI & NLP in Finance

How Natural Language Processing powers the chatbot

How NLP Works

- 1. User Input**
User types a financial question in natural language
- 2. Intent Matching**
Dialogflow's ML engine identifies the correct intent
- 3. Entity Extraction**
Key numbers like income, loan amount, tenure are extracted
- 4. Slot Filling**
Chatbot prompts for missing required parameters
- 5. Response**
Personalized financial answer is generated instantly

Key AI Concepts Used

Intent Classification

Maps user message to a predefined intent (e.g. EMI, Loan)

Named Entity Recognition

Extracts numbers, risk levels from free text

Slot Filling / Parameters

Collects required data through guided conversation

Custom Entities

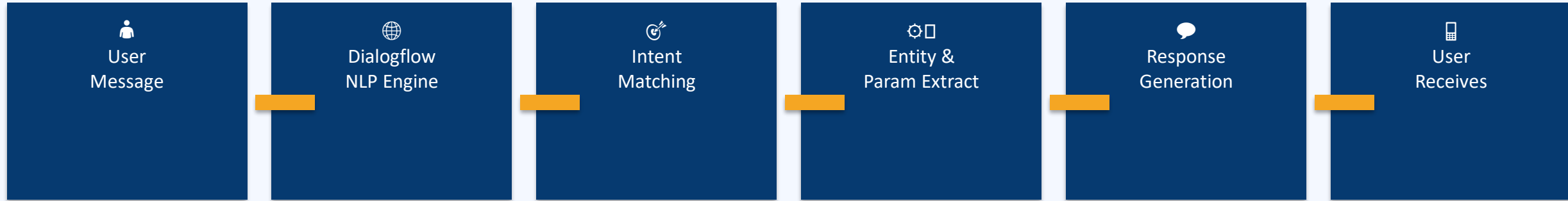
risk_level entity: LOW / MEDIUM / HIGH with synonyms

Fallback Handling

Gracefully redirects off-topic queries back to finance topics

Dialogflow Architecture

How the chatbot components connect end-to-end



Intents Configured

Default Welcome Intent
Default Fallback Intent
Loan_Eligibility_Advanced
EMI_Calculation
Investment_Advice_Advanced

Investment Planning
Low Risk Investment
Mutual Fund vs FD
Loan_Request_With_Amount

Custom Entity: risk_level

LOW	safe, conservative
MEDIUM	moderate, balanced
HIGH	aggressive, risky

Loan Eligibility Intent

Intent: Loan_Eligibility_Advanced

Training Phrases (Examples)

"I earn 60000, can I get 20 lakh home loan?"

"My salary is 70000, what loan can I get?"

"Loan eligibility for 60000 income"

"Can I get loan of 20 lakh?"

"I earn 80000 and want home loan"

Parameters Extracted

income → @sys.number — Monthly salary

Loan_amount → @sys.number — Requested loan amount

Chatbot Response Logic

▸ Accepts monthly income and desired loan amount

▸ Applies 30–40% income rule for safe EMI limit

▸ Calculates safe EMI range: $\text{income} \times 0.3$ to 0.4

▸ Assesses feasibility of requested loan amount

▸ Mentions credit score requirement: 750+

▸ Recommends longer tenure if income is borderline

 **Insert Screenshot: Loan Intent Output**

EMI Calculation Intent

Intent: EMI_Calculation

Training Phrases (Examples)

"Calculate EMI for 10 lakh at 8% for 5 years"

"EMI for 20 lakh loan at 8% interest for 60 months"

"What will be EMI for 10 lakh loan?"

"Calculate monthly installment"

"I want EMI estimate"

Parameters Extracted

LOAN_AMOUNT → @sys.number — Principal loan amount

interest → @sys.number — Annual interest rate (%)

tenure → @sys.number — Loan duration in months/years

EMI Formula (Banking Standard)

$$EMI = P \times r \times (1 + r)^n / ((1 + r)^n - 1)$$

P Principal Loan Amount (e.g. ₹10,00,000)

r Monthly Interest Rate = Annual Rate ÷ 12 ÷ 100

n Loan Tenure in Months (e.g. 60 months = 5 yrs)

Example: ₹10L at 8% for 5 years → EMI ≈ ₹20,276/month

 **Insert Screenshot: EMI Intent Output**

Investment Advice Intent

Intents: Investment_Advice_Advanced | Low Risk Investment | Mutual Fund vs FD

🔍 LOW RISK

- ▶ Fixed Deposits (FD)
- ▶ Returns: 6–7% p.a.
- ▶ Government Bonds
- ▶ Debt Mutual Funds
- ▶ Capital preservation focus

🔍 MEDIUM RISK

- ▶ Balanced Mutual Funds
- ▶ Index Funds (Nifty 50)
- ▶ Returns: 10–12% p.a.
- ▶ Moderate market exposure
- ▶ Suitable for 3–5 yr horizon

● HIGH RISK

- ▶ Equity Mutual Funds
- ▶ Direct Stock Market
- ▶ Returns: 12–18% p.a. (potential)
- ▶ High market volatility
- ▶ Suitable for 7+ yr horizon

Custom Entity: risk_level

LOW → safe, conservative, low risk

MEDIUM → moderate, balanced,
medium risk

HIGH → aggressive, high return, risky

🔗 *Insert Screenshot: Investment Intent Output*

Screenshots & Outputs

Visual evidence from Dialogflow console & chatbot testing

☞ Agent Name
& Intent List

☞ Training Phrases
(Loan Eligibility)

☞ Parameters &
Entities Config

☞ Loan Intent
Output

☞ EMI Intent
Output

☞ Investment
Intent Output

Future Improvements

How this chatbot can evolve into a full FinTech solution

Webhook Integration

Connect to live APIs for real-time loan rates, mutual fund NAV, and credit score data.

Exact EMI Calculator

Integrate webhook-based computation to deliver precise EMI amounts instantly.

Multi-Language Support

Add Hindi, Punjabi, Tamil and other Indian languages for wider reach.

Platform Integration

Deploy on WhatsApp Business, Telegram, Facebook Messenger, and mobile apps.

Bank API Linkage

Integrate with banking APIs for real loan applications and pre-approval journeys.

ML Personalization

Use past interactions to provide progressively personalized financial advice.

GitHub Repository

Project code, export files, and documentation

Repository Structure

Folder/

Dialogflow agent export (all JSON intent & entity files)

Dialogue_Flow.zip

Complete Dialogflow agent export for import/restore

README.md

Project overview, setup guide, and screenshots

Agent Name:

Lovegeet – Personal Finance AI Assistant

 [Insert GitHub URL / QR Code here](#)

To run: Import Dialogue_Flow.zip into your Dialogflow agent via the ⚙️ Settings > Export & Import tab.

Conclusion

Lovegeet – AI-Powered Personal Finance Assistant

Problem Solved

Provides instant, 24x7 financial guidance without a human advisor.

Key Capabilities

Loan eligibility assessment, EMI formula calculation, risk-based investment advice.

Scalable Design

Ready for webhook, multi-platform, and multi-language expansion.

Technology Used

Google Dialogflow NLP with custom intents, entities, and slot filling.

Impact Potential

Can serve millions of financially underserved users across India.

Learning Outcome

Demonstrates practical application of AI/NLP in the FinTech domain.

Thank You • Questions Welcome